

Git - LAB 01

1. Establish a local repository for your project with a descriptive name. Set up Git configurations by adding a username and email address. Display and modify the Git configurations, and verify if the changes are reflected in the .git/config file post-editing. Showcase the utilization of both Git Local and Global configurations.

Step 1: Establish a local repository for your project with a descriptive name.

```
root@LAPTOP-HPUOTJF1:~# mkdir DevOps_Project
root@LAPTOP-HPUOTJF1:~# cd DevOps_Project/
root@LAPTOP-HPUOTJF1:~/DevOps_Project# ls
root@LAPTOP-HPUOTJF1:~/DevOps_Project# git init
hint: Using 'master' as the name for the initial branch. This default
hint: will change to "main" in Git 3.0. To configure the initial branch
hint: to use in all of your new repositories, which will suppress this
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName
Initialized empty Git repository in /root/DevOps_Project/.git/
root@LAPTOP-HPUOTJF1:~/DevOps_Project# ls -a
.  ..  .git
```

Step 2: Set up Git configurations by adding a username and email address.

```
root@LAPTOP-HPUOTJF1:~/DevOps_Project# git config --global --list
user.name=achalpadol
user.email=achalpadol@gmail.com
```

Step 3: Display and modify the Git configurations, and verify if the changes are reflected in the .git/config file post-editing.

```
root@LAPTOP-HPUOTJF1:~/DevOps_Project# git config --global user.name "Achal S Padol"
root@LAPTOP-HPUOTJF1:~/DevOps_Project# git config --global --list
user.name=Achal S Padol
user.email=achalpadol@gmail.com
```

```
root@LAPTOP-HPUOTJF1:~/DevOps_Project# git config user.name "Achal Sunil Padol"
root@LAPTOP-HPUOTJF1:~/DevOps_Project# git config user.main "padolachal@gmail.com"
root@LAPTOP-HPUOTJF1:~/DevOps_Project# cat .git/config
[core]
    repositoryformatversion = 0
    filemode = true
    bare = false
    logallrefupdates = true
[user]
    name = Achal Sunil Padol
    main = padolachal@gmail.com
root@LAPTOP-HPUOTJF1:~/DevOps_Project# cd
```

```
root@LAPTOP-HPUOTJF1:~# cat .gitconfig
[user]
    name = Achal S Padol
    email = achalpadol@gmail.com
```

2. Set up a local repository for your project with a meaningful name. Explore the default branch in the local Git repository. Sequentially create five files, committing each with informative messages. Share the Git log output through multiple methods once the process is complete.

Step 1: Create Repo locally and sequentially create five files

```
root@LAPTOP-HPUOTJF1:~/student-repo# git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:   git config --global init.defaultBranch <name>
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:   git branch -m <name>
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /root/student-repo/.git/
root@LAPTOP-HPUOTJF1:~/student-repo#
root@LAPTOP-HPUOTJF1:~/student-repo# echo "Hello from the cloudethix" > hello.txt
root@LAPTOP-HPUOTJF1:~/student-repo# echo "Good Morning" > hello1.txt
root@LAPTOP-HPUOTJF1:~/student-repo# echo "work well" > hello2.txt
root@LAPTOP-HPUOTJF1:~/student-repo# echo "Welcome" > hello3.txt
root@LAPTOP-HPUOTJF1:~/student-repo# echo "Welcome to cloudethix" > hello4.txt
root@LAPTOP-HPUOTJF1:~/student-repo# git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    hello.txt
    hello1.txt
    hello2.txt
    hello3.txt
    hello4.txt
```

Step 2: Committing each with informative messages.

```
root@LAPTOP-HPUOTJF1:~/student-repo# git commit -m "First file is added"
[master (root-commit) 37006e0] First file is added
 1 file changed, 1 insertion(+)
 create mode 100644 hello.txt
root@LAPTOP-HPUOTJF1:~/student-repo# git add hello1.txt
root@LAPTOP-HPUOTJF1:~/student-repo# git commit -m "second file is added"
[master 87e73ad] second file is added
 1 file changed, 1 insertion(+)
 create mode 100644 hello1.txt
root@LAPTOP-HPUOTJF1:~/student-repo# git add hello2.txt
root@LAPTOP-HPUOTJF1:~/student-repo# git commit -m "Third file is added"
[master 6f4f30d] Third file is added
 1 file changed, 1 insertion(+)
 create mode 100644 hello2.txt
root@LAPTOP-HPUOTJF1:~/student-repo# git add hello3.txt
root@LAPTOP-HPUOTJF1:~/student-repo# git commit -m "Fourth file is added"
[master e9d2917] Fourth file is added
 1 file changed, 1 insertion(+)
 create mode 100644 hello3.txt
root@LAPTOP-HPUOTJF1:~/student-repo# git add hello4.txt
root@LAPTOP-HPUOTJF1:~/student-repo# git commit -m "Fifth file is added"
[master 5241284] Fifth file is added
 1 file changed, 1 insertion(+)
 create mode 100644 hello4.txt
```

Step 3: Share the Git log output through multiple methods once the process is complete.

CMD – git log

```
root@LAPTOP-HPUOTJF1:~/student-repo# git log
commit 524128417f4a8ce92eec2fcf6387937e02a59e37 (HEAD -> master)
author: Achal S Padol <achalpadol@gmail.com>
date: Fri Jul 17 05:20:53 2026 +0000

    Fifth file is added

commit e9d291756703c8e11b40e516e9c54cdd26eb77bc
author: Achal S Padol <achalpadol@gmail.com>
date: Fri Jul 17 05:20:33 2026 +0000

    Fourth file is added

commit 6f4f30df69ae6d455351800105d473565d70db1e
author: Achal S Padol <achalpadol@gmail.com>
date: Fri Jul 17 05:20:12 2026 +0000

    Third file is added

commit 87e73ad412258ad0ce76aa4800cebf8b6142b068
author: Achal S Padol <achalpadol@gmail.com>
date: Fri Jul 17 05:19:28 2026 +0000

    second file is added

commit 37006e0c6c365163caeaf4cd4dc7befc1dc21409
author: Achal S Padol <achalpadol@gmail.com>
date: Fri Jul 17 05:18:57 2026 +0000

    First file is added
```

CMD – git log –oneline

```
root@LAPTOP-HPUOTJF1:~/student-repo# git log --oneline
5241284 (HEAD -> master) Fifth file is added
e9d2917 Fourth file is added
6f4f30d Third file is added
87e73ad second file is added
37006e0 First file is added
```

CMD – git log --graph

```
root@LAPTOP-HPUOTJF1:~/student-repo# git log --graph
* commit 524128417f4a8ce92eec2fcf6387937e02a59e37 (HEAD -> master)
  Author: Achal S Padol <achalpadol@gmail.com>
  Date:   Fri Jul 17 05:20:53 2026 +0000

    Fifth file is added

* commit e9d291756703c8e11b40e516e9c54cdd26eb77bc
  Author: Achal S Padol <achalpadol@gmail.com>
  Date:   Fri Jul 17 05:20:33 2026 +0000

    Fourth file is added

* commit 6f4f30df69ae6d455351800105d473565d70db1e
  Author: Achal S Padol <achalpadol@gmail.com>
  Date:   Fri Jul 17 05:20:12 2026 +0000

    Third file is added

* commit 87e73ad412258ad0ce76aa4800cebf8b6142b068
  Author: Achal S Padol <achalpadol@gmail.com>
  Date:   Fri Jul 17 05:19:28 2026 +0000

    second file is added

* commit 37006e0c6c365163caeaf4cd4dc7befc1dc21409
  Author: Achal S Padol <achalpadol@gmail.com>
  Date:   Fri Jul 17 05:18:57 2026 +0000

    First file is added
```

CMD - git log --oneline --graph

```
root@LAPTOP-HPUOTJF1:~/student-repo# git log --oneline --graph
* 5241284 (HEAD -> master) Fifth file is added
* e9d2917 Fourth file is added
* 6f4f30d Third file is added
* 87e73ad second file is added
* 37006e0 First file is added
```

CMD – git log –stat

```
root@LAPTOP-HPUOTJF1:~/student-repo# git log --stat
commit 524128417f4a8ce92eec2fcf6387937e02a59e37 (HEAD -> master)
Author: Achal S Padol <achalpadol@gmail.com>
Date:   Fri Jul 17 05:20:53 2026 +0000

    Fifth file is added

    hello4.txt | 1 +
    1 file changed, 1 insertion(+)

commit e9d291756703c8e11b40e516e9c54cdd26eb77bc
Author: Achal S Padol <achalpadol@gmail.com>
Date:   Fri Jul 17 05:20:33 2026 +0000

    Fourth file is added

    hello3.txt | 1 +
    1 file changed, 1 insertion(+)

commit 6f4f30df69ae6d455351800105d473565d70db1e
Author: Achal S Padol <achalpadol@gmail.com>
Date:   Fri Jul 17 05:20:12 2026 +0000

    Third file is added

    hello2.txt | 1 +
    1 file changed, 1 insertion(+)

commit 87e73ad412258ad0ce76aa4800cebf8b6142b068
Author: Achal S Padol <achalpadol@gmail.com>
Date:   Fri Jul 17 05:19:28 2026 +0000

    second file is added

    hello1.txt | 1 +
    1 file changed, 1 insertion(+)

commit 37006e0c6c365163caeaf4cd4dc7befc1dc21409
Author: Achal S Padol <achalpadol@gmail.com>
Date:   Fri Jul 17 05:18:57 2026 +0000

    first file is added
```

CMD – git log –all –decorate –oneline --graph

```
root@LAPTOP-HPUOTJF1:~/student-repo# git log --all --decorate --oneline --graph
* 5241284 (HEAD -> master) Fifth file is added
* e9d2917 Fourth file is added
* 6f4f30d Third file is added
* 87e73ad second file is added
* 37006e0 First file is added
```

3. Open a GitHub account and initiate a remote Git repository. Examine the default branch in the remote repository. Create and commit the following files one by one with meaningful commit messages:

i. README.md

ii. send_disk_alert.sh

iii. create_users.sh

iv. check_service.sh.

Step 1: Open a GitHub account and initiate a remote Git repository.

Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).
Required fields are marked with an asterisk (*).

1 General

Owner * achalpadol / **Repository name *** remote-repo
✔ remote-repo is available.

Great repository names are short and memorable. How about [turbo-octo-lamp?](#)

Description

0 / 350 characters

2 Configuration

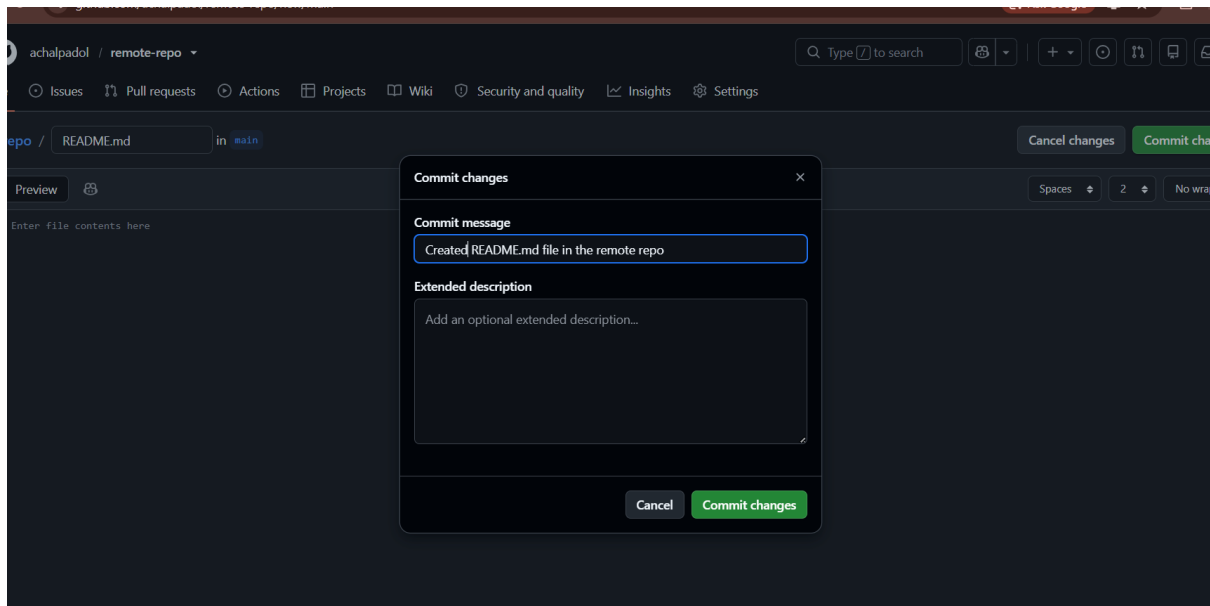
Choose visibility *
Choose who can see and commit to this repository. **Public**

Start with a template
Templates pre-configure your repository with files. **No template**

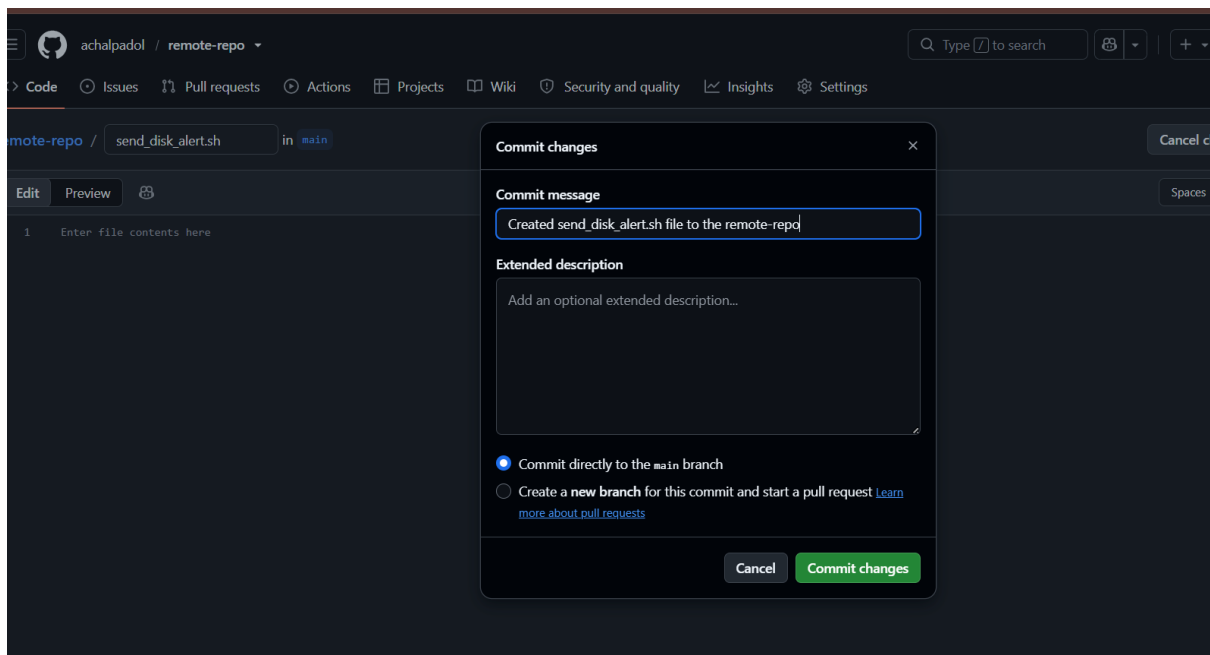
Add README
READMEs can be used as longer descriptions. [About READMEs](#). **Off**

Step 2: Create and commit the following files one by one with meaningful commit messages:

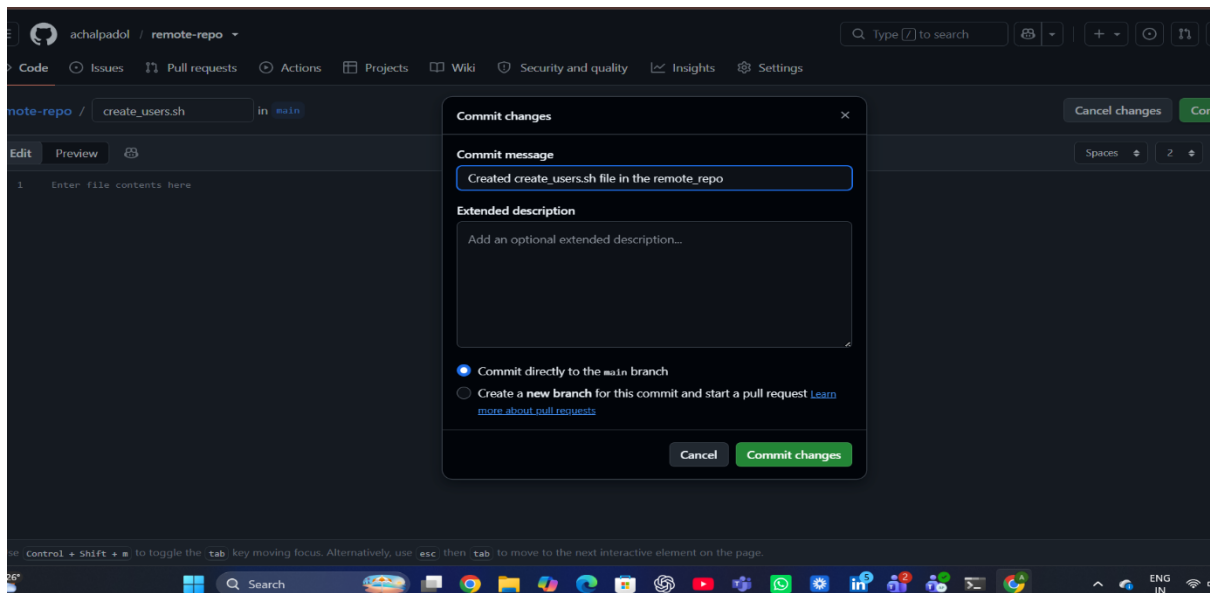
Step 1: Commit the File README.md



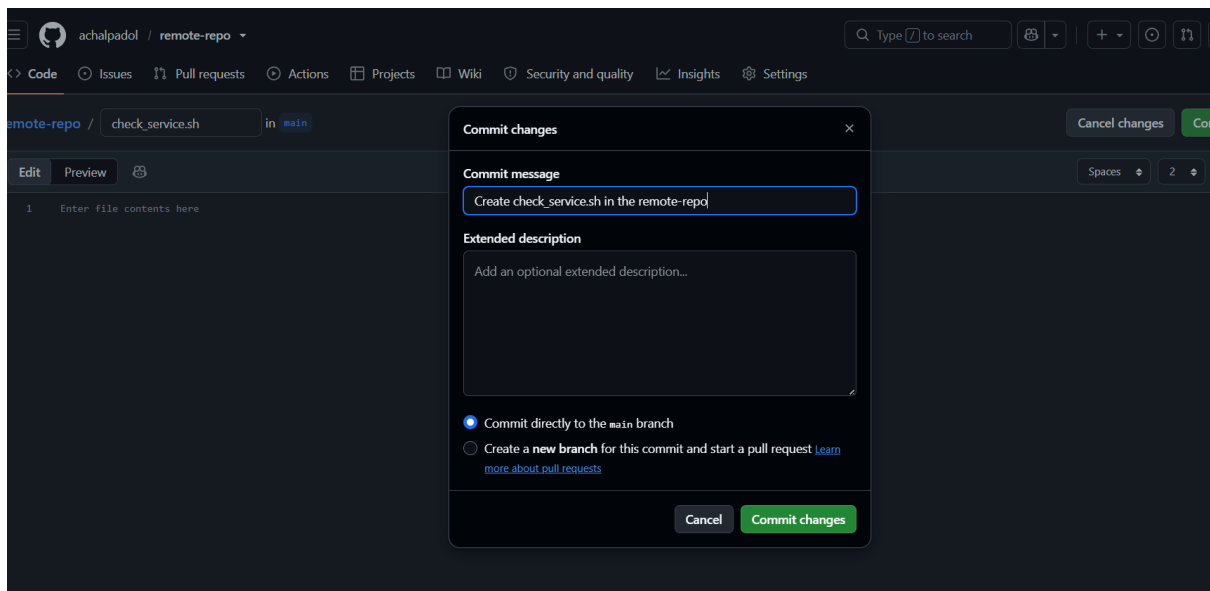
Step 2: Commit the send_disk_alert.sh

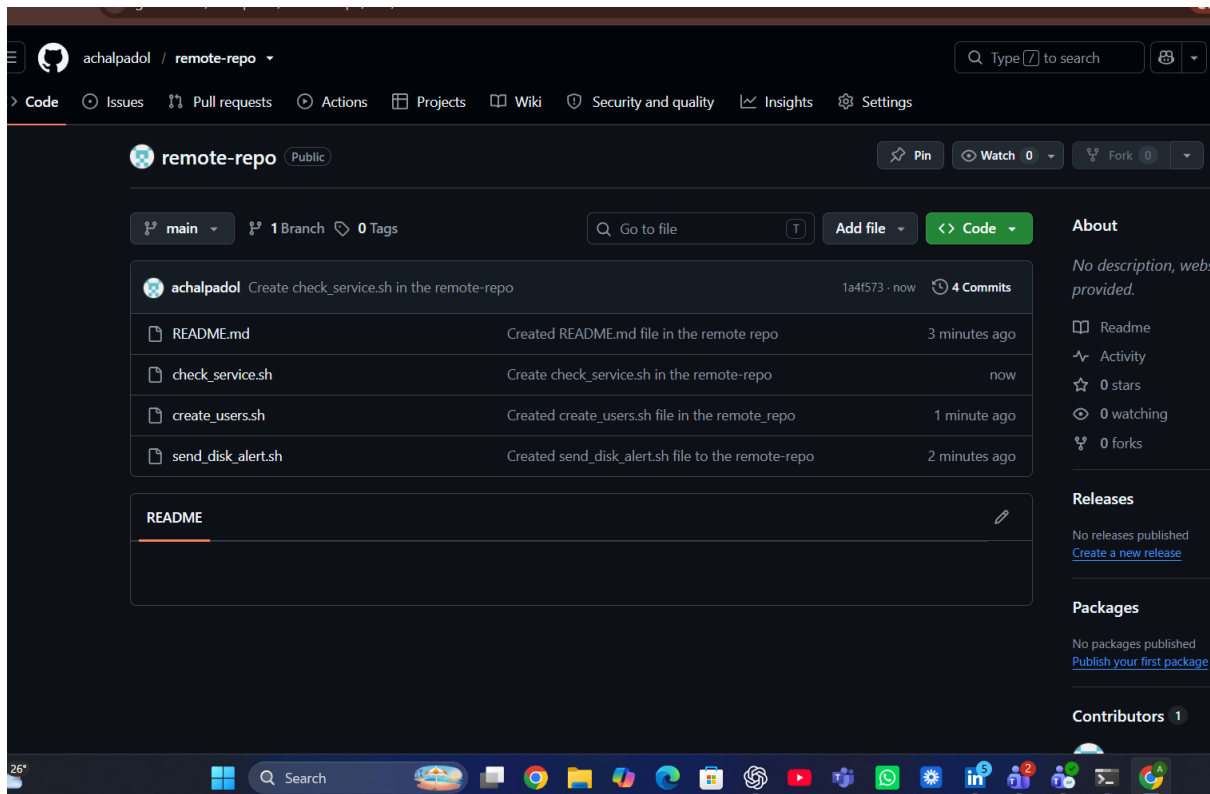


Step 3: Commit the create_users.sh



Step 4: Commit the check_service.sh.





4. Establish a local repository for your project with a descriptive name. Illustrate the implementation of a Git-Ignore file by committing multiple files to the local repository.

Step 1: Establish a local repository for your project with a descriptive name.

```
oot@LAPTOP-HPUOTJF1:~/GitIgnore-Repo# cd
oot@LAPTOP-HPUOTJF1:~# cd GitIgnore-Repo/
oot@LAPTOP-HPUOTJF1:~/GitIgnore-Repo# git init
int: Using 'master' as the name for the initial branch. This default branch name
int: will change to "main" in Git 3.0. To configure the initial branch name
int: to use in all of your new repositories, which will suppress this warning,
int: call:
int:
int:   git config --global init.defaultBranch <name>
int:
int: Names commonly chosen instead of 'master' are 'main', 'trunk' and
int: 'development'. The just-created branch can be renamed via this command:
int:
int:   git branch -m <name>
int:
int: Disable this message with "git config set advice.defaultBranchName false"
initialized empty Git repository in /root/GitIgnore-Repo/.git/
oot@LAPTOP-HPUOTJF1:~/GitIgnore-Repo# ls
pp.js index.html style.css
oot@LAPTOP-HPUOTJF1:~/GitIgnore-Repo# git status
n branch master

o commits yet

ntracked files:
(use "git add <file>..." to include in what will be committed)
.gitignore
app.js
index.html
style.css
```

Step 2: Illustrate the implementation of a Git-Ignore file by committing multiple files to the local repository.

```
root@LAPTOP-HPU0TJF1:~/GitIgnore-Repo# git add .
root@LAPTOP-HPU0TJF1:~/GitIgnore-Repo# git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   .gitignore
    new file:   app.js
    new file:   index.html
    new file:   style.css

root@LAPTOP-HPU0TJF1:~/GitIgnore-Repo# git commit -m "Files are added"
[master (root-commit) c6da3ea] Files are added
 4 files changed, 4 insertions(+)
 create mode 100644 .gitignore
 create mode 100644 app.js
 create mode 100644 index.html
 create mode 100644 style.css
root@LAPTOP-HPU0TJF1:~/GitIgnore-Repo# git status
On branch master
nothing to commit, working tree clean
root@LAPTOP-HPU0TJF1:~/GitIgnore-Repo# git status --ignored
On branch master
Ignored files:
  (use "git add -f <file>..." to include in what will be committed)
    .env
```

```
root@LAPTOP-HPU0TJF1:~/GitIgnore-Repo# git log
commit c6da3ea2c93526d973ef028057ffe8997a684123 (HEAD -> master)
Author: Achal S Padol <achalpadol@gmail.com>
Date:   Fri Jul 17 05:56:51 2026 +0000

    Files are added
root@LAPTOP-HPU0TJF1:~/GitIgnore-Repo# ls -a
.  ..  .env  .git  .gitignore  app.js  index.html  style.css
root@LAPTOP-HPU0TJF1:~/GitIgnore-Repo# |
```

5. Create a local repository with a meaningful project name. Demonstrate the utilization of GIT working area, staging area, and commit area. Experiment with committing, deleting, and restoring files from the repository. Additionally, add files to the staging area and restore them from the staging area.

Step 1: Create a local repository with a meaningful project name.

```

root@LAPTOP-HPUOTJF1:~# mkdir GitWorking
root@LAPTOP-HPUOTJF1:~# cd GitWorking/
root@LAPTOP-HPUOTJF1:~/GitWorking# git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"

```

Step 2: Demonstrate the utilization of GIT working area, staging area, and commit area.

```

root@LAPTOP-HPUOTJF1:~/GitWorking# touch file1
root@LAPTOP-HPUOTJF1:~/GitWorking# git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
       file1

nothing added to commit but untracked files present (use "git add" to track)
root@LAPTOP-HPUOTJF1:~/GitWorking# git add file1
root@LAPTOP-HPUOTJF1:~/GitWorking# git commit -m "File1 is added"
[master (root-commit) 6d150d0] File1 is added
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 file1
root@LAPTOP-HPUOTJF1:~/GitWorking# git log
commit 6d150d0b449f36055ec1d7dd3590dca34ae10bfff (HEAD -> master)
Author: Achal S Padol <achalpadol@gmail.com>
Date:   Fri Jul 17 06:03:49 2026 +0000

```

Step 3: Experiment with committing, deleting, and restoring files from the repository.

```

Date:   Fri Jul 17 06:03:49 2026 +0000

File1 is added
root@LAPTOP-HPUOTJF1:~/GitWorking# ls
file1
root@LAPTOP-HPUOTJF1:~/GitWorking# rm -f file1
root@LAPTOP-HPUOTJF1:~/GitWorking# ls
root@LAPTOP-HPUOTJF1:~/GitWorking# git restore file1
root@LAPTOP-HPUOTJF1:~/GitWorking# ls
file1

```

Step 4: Additionally, add files to the staging area and restore them from the staging area.

```
root@LAPTOP-HPUOTJF1:~/GitWorking# git add file1
root@LAPTOP-HPUOTJF1:~/GitWorking# git status
On branch master
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   file1

root@LAPTOP-HPUOTJF1:~/GitWorking# git restore --staged file1
root@LAPTOP-HPUOTJF1:~/GitWorking# git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   file1
```